VOLUME 6 — MODEL PAPER 1 (Difficulty: Easy → Moderate)

NID DAT (B.Des) Prelims: Practice Model Paper 1
Format follows the scheme printed in the NID Admissions Handbook: Part I (22 objective) + Part II (4 subjective). This is a practice paper written for this package. It is not an official NID paper.

INSTRUCTIONS

1. Time: **3 hours (180 minutes)**. Total marks: **100**. No negative marking.
2. **Part I** (Q1–22): objective. Q1–14 carry 1 mark each; Q15–22 carry 2 marks each. Total 30.
3. **Part II** (Q23–26): subjective. Q23 = 15, Q24 = 20, Q25 = 15, Q26 = 20. Total 70.
4. Answer Part II in the space given (use an A4 sheet per question when practising at home).
5. Allowed: pencils, eraser, sharpener, scale, colour pencils, crayons or oil pastels, sketch pens.
6. Suggested timing: Part I = 25 min · Q23 = 28 min · Q24 = 38 min · Q25 = 28 min · Q26 = 38 min · buffer = 23 min.

PART I — OBJECTIVE (30 marks)

Q1–Q14 (1 mark each)

Q1. On the painter's (RYB) colour wheel, the complementary colour of red is:
(a) Blue (b) Green (c) Yellow (d) Violet

Q2. How many letters of the word **TAXI** look exactly the same in a vertical mirror?
(a) 1 (b) 2 (c) 3 (d) 4

Q3. Warli painting is a tribal art form of:
(a) Bihar (b) Maharashtra (c) Odisha (d) Madhya Pradesh

Q4. How many faces does a triangular prism have?
(a) 4 (b) 5 (c) 6 (d) 9

Q5. Which company's logo hides an arrow in its negative space?
(a) Apple (b) Nike (c) FedEx (d) Pepsi

Q6. Odd one out: Chair, Table, Sofa, Spoon
(a) Chair (b) Table (c) Sofa (d) Spoon

Q7. A "tint" of a colour is produced by adding:
(a) black (b) grey (c) white (d) its complement

Q8. The Gateway of India is located in:
(a) Delhi (b) Kolkata (c) Mumbai (d) Chennai

Q9. Which regular shape can tile a floor by itself with no gaps?
(a) Circle (b) Regular pentagon (c) Regular hexagon (d) Regular octagon

Q10. 2, 4, 8, 16, ?
(a) 24 (b) 32 (c) 30 (d) 64

Q11. A straight railway track drawn going away to the horizon is best shown using:
(a) one-point perspective (b) two-point perspective (c) isometric view (d) plan view

Q12. A square sheet is folded in half and then in half again. One hole is punched through all layers, away from the folds. How many holes appear when it is unfolded?
(a) 2 (b) 3 (c) 4 (d) 8

Q13. Kanjeevaram silk saris come from:
(a) Karnataka (b) Tamil Nadu (c) Kerala (d) Andhra Pradesh

Q14. Which of these is a sans-serif typeface?
(a) Times New Roman (b) Garamond (c) Helvetica (d) Baskerville

Q15–Q22 (2 marks each)

Q15. A 4 × 4 × 4 cube is painted on all faces and cut into 64 unit cubes. How many unit cubes have exactly ONE face painted?
(a) 8 (b) 16 (c) 24 (d) 32

Q16. A wall clock shows 7:20. What time does its image in a vertical mirror show?
(a) 4:40 (b) 5:40 (c) 4:20 (d) 3:40

Q17. Five posters P, Q, R, S, T hang in a row. R is in the middle. P is at the left end. Q is immediately to the right of R. T is not next to P. Which poster is second from the left?
(a) Q (b) S (c) T (d) R

Q18. In the cube net below, which face is opposite face 2?

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   [1]
[2][3][4][5]
   [6]
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(a) 3 (b) 4 (c) 5 (d) 6

Q19. A room 3 m × 2.4 m is to be tiled with 30 cm × 30 cm square tiles. How many tiles are needed?
(a) 64 (b) 72 (c) 80 (d) 96

Q20. At a busy bus stop, elderly passengers cannot read the small route numbers on approaching buses. Which solution is MOST effective?
(a) Print a timetable booklet
(b) Put large, high-contrast route numbers on the bus front and sides, with an audio announcement at the stop
(c) Paint the buses in brighter colours
(d) Ask passengers to carry binoculars

Q21. A painting shows a single red apple among many green apples. The design principle MOST clearly used is:
(a) Rhythm (b) Emphasis through contrast (c) Symmetry (d) Proportion

Q22. Four gears A, B, C, D are in a line, each meshing with the next. If A turns clockwise, D turns:
(a) clockwise (b) anticlockwise (c) not at all (d) alternately

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PART II — SUBJECTIVE (70 marks)

Q23. Memory Drawing (15 marks)
Draw a roadside **tea stall (chai tapri)** from memory, with the tea-seller and three customers. Your drawing must include **at least eight objects** that make the place recognisable. Use pencil shading; colour is optional.

Q24. Sequential Drawing (20 marks)
In **four frames**, show how a small seed planted by a child grows over the years into a tree that gives shade to a bus stop. Show the passing of time clearly. Keep the same location in all frames.

Q25. Poster Design (15 marks)
Design a poster on **"Save Electricity"** for a school notice board. Use **not more than three colours** (black and white count as colours if used). Include a short headline of your own.

Q26. Problem Solving (20 marks)
Children in your housing colony play cricket in the parking area, and cars, windows and children often get hurt or damaged.
(a) List **three** problems in this situation (3 marks).
(b) Design **one** solution and present it as a labelled sketch (14 marks).
(c) Write **two lines** explaining how your solution helps everyone involved (3 marks).

ANSWER KEY — PART I

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	b	7	c	13	b	19	c
2	d	8	c	14	c	20	b
3	b	9	c	15	c	21	b
4	b	10	b	16	a	22	b
5	c	11	a	17	b		
6	d	12	c	18	b		

DETAILED SOLUTIONS — PART I

Q1 → (b) Green. On the RYB wheel red sits opposite green. *Why not blue?* Blue is a primary colour 120° away (the red–yellow–blue triad). *Concept:* complements · *Diff:* E · *Skill:* colour theory.

Q2 → (d) 4. T, A, X and I are all symmetric about a vertical axis, so each looks the same in a vertical mirror. (The word as a whole reads "IXAT", but the question asks about the letters.) *Concept:* mirror symmetry · *Diff:* E.

Q3 → (b) Maharashtra. Warli comes from the Thane–Palghar region and is painted in white rice paste on mud walls. Bihar = Madhubani; Odisha = Saura and Pattachitra; MP = Gond. *Concept:* folk art · *Diff:* E.

Q4 → (b) 5. Two triangles + three rectangles = 5 faces. It has 9 edges (option d is the trap) and 6 vertices. *Concept:* solids · *Diff:* E.

Q5 → (c) FedEx. The arrow is formed between the "E" and the "x". *Concept:* negative space · *Diff:* E.

Q6 → (d) Spoon. It is cutlery; the rest are furniture. *Concept:* classification · *Diff:* E.

Q7 → (c) white. Tint = hue + white; shade = + black; tone = + grey. *Concept:* colour terms · *Diff:* E.

Q8 → (c) Mumbai. Designed by George Wittet; completed 1924. *Concept:* GK · *Diff:* E.

Q9 → (c) Regular hexagon. Its interior angle of 120° fits exactly 3 times around a point (360°). Pentagons (108°) and octagons (135°) leave gaps; circles always leave gaps. *Concept:* tessellation · *Diff:* E.

Q10 → (b) 32. Each term doubles. *Concept:* series · *Diff:* E.

Q11 → (a) one-point. The rails recede to a single vanishing point on the horizon. *Concept:* perspective · *Diff:* E.

Q12 → (c) 4. Two folds make $2^2 = 4$ layers, so 4 holes. *Concept:* paper folding · *Diff:* E.

Q13 → (b) Tamil Nadu (Kanchipuram). Karnataka = Mysore silk. *Concept:* textiles · *Diff:* E.

Q14 → (c) Helvetica. The other three are serif typefaces. *Concept:* typography · *Diff:* E.

Q15 → (c) 24. One-face cubes = $6(n-2)^2 = 6 \times 2^2 = 24$. Check: corners 8 + edges 12 × 2 = 24 + faces 24 + inside 8 = 64 ✓. *Concept:* painted cube · *Diff:* M.

Q16 → (a) 4:40. 11:60 − 7:20 = 4:40. *Check:* the minute hand at "4" (20 min) mirrors to "8" (40 min); the hour hand just past 7 mirrors to just before 5. So 4:40 ✓. *Concept:* mirror clock · *Diff:* M.

Q17 → (b) S. Positions 1–5: P = 1, R = 3, Q = 4. Positions 2 and 5 are left for S and T. T is not next to P, so T ≠ 2. Therefore T = 5 and S = 2. *Concept:* arrangement · *Diff:* M.

Q18 → (b) 4. In the row 2-3-4-5, faces with one face between them are opposite: 2–4 and 3–5. Faces 1 and 6, attached above and below face 3, are opposite each other. *Concept:* nets · *Diff:* M.

Q19 → (c) 80. 300 ÷ 30 = 10 tiles along the length; 240 ÷ 30 = 8 tiles along the width; 10 × 8 = 80. *Concept:* area · *Diff:* M.

Q20 → (b). It fixes the root cause (legibility at a distance) and adds a non-visual channel (audio) for people with poor eyesight. (a) does not help identify an arriving bus, (c) does not identify the route, and (d) is absurd. *Concept:* inclusive design · *Diff:* M.

Q21 → (b) Emphasis through contrast. The one different colour becomes the focal point. *Concept:* principles · *Diff:* E.

Q22 → (b) anticlockwise. Meshing gears alternate: A clockwise, B anticlockwise, C clockwise, D anticlockwise. *Concept:* mechanical reasoning · *Diff:* M.

DETAILED SOLUTIONS — PART II (model approaches + marking rubrics)

> These rubrics are **practice benchmarks** built from the inferred NID criteria (Vol 1 §4). They are not NID's official scheme.

Q23. Chai tapri from memory (15 marks)

What a strong answer includes

- **Setting:** a small wooden or metal stall on a footpath, sometimes under a tree, with a tarpaulin or tin roof.
- **The 8+ identifying objects:**
    1. kerosene/LPG stove or angithi
    2. aluminium or steel kettle (patila)
    3. strainer
    4. glass jars of rusk, biscuits and toast
    5. small glass tumblers or kulhads
    6. sugar and tea-leaf tins
    7. milk vessel
    8. ginger and grater
    9. bench or stools
    10. hanging packets of snacks/chips
    11. a small radio
    12. a handwritten price board ("Chai ₹10")
    13. dustbin
    14. bicycle parked nearby
- **People:** the vendor pouring tea from a height (a classic action); one customer sipping, one reading a newspaper, one waiting with coins.
- **Composition:** a 2-point view of the stall corner with the vendor as the focal point (at a thirds point), and steam rising from the kettle. Customers overlap to show depth.

Step-by-step approach (28 min)

1. 2 min: list the objects in the margin.
2. 4 min: horizon line, then the stall block in 2-point perspective.
3. 8 min: gesture figures in correct proportion against the stall (counter height about 0.9 m, roughly the waist of an adult).
4. 10 min: details and objects; label them lightly if there is any doubt.
5. 4 min: shading (light from one side), steam, ground shadow.

Marking rubric (15)

Criterion	Marks
Recognisability, 8+ authentic objects	4
People: proportion, actions, interaction	3
Perspective & composition	3
Observation detail and memory accuracy	3
Rendering: line quality, shading	2

Common mistakes: a generic shop with no tea-specific objects; people taller than the stall roof; leaving out the activity (an empty stall).

Q24. Seed to shade tree, 4 frames (20 marks)

Model plan (fixed camera position, same bus-stop location)

Frame	Time	Content
1	Year 0	A child (about 8) plants a sapling beside a bare bus-stop pole on a hot day. A parent waters it. The sun is harsh; commuters squint.

Frame	Time	Content
2	Year 3	A sapling with a tree guard. The child (now about 11) waters it with a school bottle. A small bench has been added.
3	Year 8	A young tree as tall as the stop sign. The teenager chats with friends under thin shade. Birds arrive.
4	Year 15	A large tree spreading shade over the whole stop. Commuters, a vendor and elderly people rest in the shade. The young adult (now about 23) passes by and smiles. A nest is in the tree.

Showing the passing of time:

- the child ages (height, clothes: school uniform → college bag → office wear)
- the tree grows
- seasonal cues
- the surroundings change (a newer bus model, a new building in the background)
- optionally, small captions: "Year 0, 3, 8, 15"

Marking rubric (20)

Criterion	Marks
Clear narrative and the passage of time	5
Continuity (same location and camera, recognisable character)	4
Drawing quality: proportions of people and tree	4
Composition of each frame	3
Emotional or social message (shade = care for community)	2
Completion & neatness	2

Common mistakes: changing the viewpoint so the location becomes unrecognisable; the tree growing but the child staying the same; empty frames without people.

Q25. "Save Electricity" poster, ≤ 3 colours (15 marks)

Idea options (choose one and execute it boldly)

1. **Visual metaphor:** a light bulb drawn as an hourglass, with the "sand" (light) running out. Headline: "**Light is running out. Switch off.**"
2. **Switch as hero:** a giant switch in the OFF position casts a shadow shaped like a planet Earth smiling. Headline: "**One click. One planet.**"
3. **Classroom cue:** an empty classroom with fans and lights on, labelled "Nobody's here... except the bill." Headline: "**Last one out? Lights out.**"

Execution (for idea 1)

- **Colours:** yellow (the light), black (outlines and text) and white (the background).
- **Hierarchy:** the headline at the top (largest), the bulb-hourglass in the middle (dominant visual), and a small tip line below: "Switch off fans & lights when you leave the room."
- **Composition:** central, symmetrical, with plenty of negative space for impact.

Marking rubric (15)

Criterion	Marks
Concept originality & clarity	5
Hierarchy & layout	3

Criterion	Marks
Typography / lettering	3
Colour constraint followed effectively	2
Overall impact & neatness	2

Common mistakes: more than 3 colours (an instruction violation); too much text; the cliché of a crying Earth with nothing new added.

Q26. Cricket in the parking area (20 marks)

(a) Three problems (3 marks, 1 each): any three of:

- balls hit cars and windows, causing damage
- children are at risk from moving cars
- there is no designated play space
- noise disturbs residents
- conflict between parents and car owners
- the elderly face falling risks from flying balls

(b) Model solution (14 marks): "Net-Pitch Corner"

- Convert a little-used corner (or a rooftop or terrace) into a **netted cricket cage**: a frame of GI pipes with nylon netting on all sides and the top, a **concrete pitch strip** with a mat, a soft-rubber **ball-only rule**, and **timing slots** (5–7 pm) displayed on a signboard.
- Parking lines are moved to separate the play zone with **bollards and a painted pedestrian path**.
- A **rebound wall** inside lets solo players practise.
- **Labels on the sketch:**
  - 3 m high netting
  - entry gate with spring hinge
  - soft-ball rule board
  - bench for parents
  - LED light for evening
  - pitch mat
  - bollards
- **Present as:** an isometric or 2-point sketch of the cage in the parking area, plus a small plan view showing the zones (play / parking / walkway).

(c) Two lines (3 marks): "The netted cage keeps balls inside, so cars and windows are safe, and the bollards keep cars out of the play zone, so children are safe. Fixed play timings and soft balls reduce noise and conflict, so both residents and children win."

Marking rubric for (b)

Criterion	Marks
Solution addresses all 3 stated problems	4
Feasibility (cost, space, materials)	3
Clarity of sketch & labels	4
User consideration (children, parents, car owners, elderly)	2
Originality	1

Common mistakes: "ban cricket" (not design thinking); an app-based solution with no physical change; an unlabelled sketch.

SCORING INTERPRETATION — MODEL PAPER 1

Practice benchmarks only. These are NOT NID cutoffs (NID does not publish verifiable cutoffs, and Prelims performance only shortlists for Mains).

Total score	Band	Meaning	Next step
80–100	**Excellent**	Strong command of easy–moderate material	Move to Papers 3–5; push originality
65–79	**Strong**	Solid base; a few gaps	Review mistakes; do Paper 2 within 5 days
50–64	**Good**	Competitive, but drawing or reasoning needs work	Daily scene drawing + Vol 4B drills
35–49	**Needs improvement**	Clear weaknesses in one or more areas	Revisit Vol 2A & 3 basics; retake in 10 days
< 35	**Major improvement required**	Fundamentals missing	Follow the 90-day plan from Day 1 (Vol 13)

Section diagnostics: Part I below 20/30 means drill visual reasoning. Any subjective question below 50% of its marks means repeat a similar prompt from Vol 3 §13 within 3 days.

NEXT VOLUME: Volume 7 — Model Paper 2 (`07-VOL7-MODEL-PAPER-2.md`)